

Aquifer Model

There are two aquifer models available in GEM. These are the steady-state aquifer and the Carter-Tracy aquifer model ([Dake, 1978](#)).

Steady-State Aquifer

This model assumes that the pressure at the external boundary of the aquifer does not change. Then the rate of water influx is

$$Q_A = C_A \left(p_A^{(i)} - \bar{p} \right) \quad (\text{A4.1})$$

where

C_A = total compressibility of the aquifer = water compressibility + rock compressibility

$p_A^{(i)}$ = pressure at the external boundary of the aquifer and is constant

$\bar{p}$ = the average pressure at the aquifer-reservoir interface

Carter-Tracy Aquifer

In this approach the rate of water influx during the current timestep is assumed to be given by ([Carter and Tracy, 1960](#)).

$$q_w^{n+1} = a(t) + b(t) \left\{ p_{ij}^{n+1} - p_{ij}^n \right\}$$

where p_{ij}^n is the pressure of block i, j at timestep n , and

$$a = \left[\frac{B_l \left(p_{ij}^o - p_{ij}^n \right) - W^n P I'^{n+1}}{P I^{n+1} - t_D^n P I'^{n+1}} \right] \left[\frac{t_D^{n+1} - t_D^n}{t^{n+1} - t^n} \right]$$

$$b = \left[\frac{-B_l}{PI^{n+1} - t_D^n PI'^{n+1}} \right] \left[\frac{t_D^{n+1} - t_D^n}{t^{n+1} - t^n} \right]$$

W = cumulative water influx

$$B_l = 2 \pi f \phi h \bar{c} r_o^2$$

h = aquifer thickness

r_o = effective reservoir radius

$\bar{c}$ = total compressibility (water + rock)

f = $\theta/2\pi$ angle of influence

$$t_D = \text{dimensionless time} = \frac{kt}{\phi \mu \bar{c} r_o^2}$$

$PI(t_D)$ = dimensionless water influx function for constant terminal rate case

$PI'(t_D)$ = derivative of $PI(t_D)$ with respect to t_D

θ = encroachment angle

Further details of the Carter-Tracy unsteady-state aquifer model can be found in [Carter and Tracy_\(1960\)](#). The built-in aquifer influx function PI is shown in Table A-1. This influx function is only valid for a radial aquifer.

Table A-1: Built in Aquifer Influx Function

t	PI
0.00	0.000
0.01	0.112
0.05	0.229
0.10	0.315
0.15	0.376
0.20	0.424
0.30	0.503
0.50	0.616
0.70	0.702
1.00	0.802
1.50	0.927
2.00	1.020
3.00	1.169
5.00	1.362
7.00	1.500
10.00	1.651
15.00	1.829
20.00	1.960
30.00	2.147
40.00	2.282
50.00	2.388
60.00	2.476
70.00	2.550
80.00	2.615
90.00	2.672
100.0	2.723
200.0	3.064

t	PI
300.0	3.263
400.0	3.406
500.0	3.516
600.0	3.608
700.0	3.684
800.0	3.750
900.0	3.809
1000.0	3.860